A Reproducible Framework for Vietnamese Scene-Text Recognition

Data preparation, text detection, sequence recognition, and evaluation

Independent Research Manuscript

Prepared for educational and portfolio use

Abstract

Vietnamese optical character recognition (OCR) is challenging because the writing system combines Latin characters with multiple diacritics, while practical images often contain blur, perspective distortion, uneven illumination, and varied typography. This paper presents a reproducible framework for developing a two-stage Vietnamese scene-text OCR pipeline. The framework covers text detection, sequence recognition, Unicode-aware label handling, dataset construction, augmentation, evaluation, and error analysis. It discusses segmentation-based detection methods such as Differentiable Binarization and recognition approaches based on Connectionist Temporal Classification. Particular emphasis is placed on preventing data leakage, preserving small diacritic marks during preprocessing, and reporting character-level as well as word-level errors. The manuscript is methodological: it proposes an experimental protocol and does not claim new benchmark results. Its goal is to provide a practical foundation for transparent, extensible OCR experiments.

Keywords: optical character recognition; Vietnamese; scene text; text detection; CTC; Unicode; computer vision

1. Introduction

Optical character recognition (OCR) converts text in images into machine-readable sequences. Modern OCR systems commonly separate the task into text detection, which locates text regions, and text recognition, which predicts the characters within those regions. Although OCR has become highly effective for many high-resource languages, Vietnamese introduces practical challenges that are not fully captured by generic English-language benchmarks.

Vietnamese uses Latin letters together with diacritics that indicate vowel quality and tone. A single base letter can appear in several visually similar forms, and small marks may be blurred, merged with neighboring strokes, or lost during resizing and compression. These difficulties are especially relevant for photographs of signs, forms, receipts, and documents captured with mobile devices under uncontrolled lighting.

This paper outlines a reproducible development framework for Vietnamese scene-text OCR. It focuses on data representation, detection and recognition stages, preprocessing, evaluation, and error analysis. The framework is intended for small research projects and educational implementations rather than as a claim that a new model has achieved state-of-the-art results.

2. Vietnamese OCR Challenges

2.1 Character inventory and visual similarity

A Vietnamese OCR system must represent digits, punctuation, uppercase and lowercase letters, and the full set of accented vowel forms. A character inventory should be defined explicitly and version-controlled. Unicode normalization is important because visually identical text can be encoded using composed characters or a base character followed by combining marks. Training labels and evaluation scripts should apply a consistent normalization policy.

2.2 Image quality and layout variation

Real-world images vary in perspective, blur, illumination, background texture, font, scale, and orientation. Text may be printed, handwritten, curved, partially occluded, or arranged in multiple lines. A training set dominated by clean document scans will not necessarily generalize to street signs or mobile photographs. Dataset splits should be made at the document or source-image level to reduce leakage between training and test sets.

3. A Two-Stage OCR Pipeline

3.1 Text detection

The detection stage predicts text regions, often as bounding boxes or polygons. Segmentation-based methods estimate a text probability map and then derive boundaries through post-processing. Differentiable Binarization (DB) is one representative approach: it learns a probability map and an adaptive threshold map, allowing the binarization process to be optimized jointly with the network. Polygon annotations can preserve the shape of rotated or irregular text regions more accurately than axis-aligned boxes.

For a compact educational implementation, the detection pipeline should be divided into image loading, annotation parsing, target-map generation, augmentation, model inference, and polygon post-processing. Each stage should be tested independently. Annotation conversion bugs can create apparently unstable training even when the neural network implementation is correct.

3.2 Text recognition

The recognition stage maps a cropped text image to a character sequence. A convolutional recurrent architecture with Connectionist Temporal Classification (CTC) is a common baseline because it can learn alignments without requiring a character-level segmentation for every image. Attention-based encoder-decoder models provide another option, particularly when sequence context is useful, but they may require more careful decoding and training design.

The output vocabulary must match the label encoding. If a blank token is used by CTC, its index must be reserved and excluded from the ordinary character inventory. The mapping from Unicode strings to token IDs should be deterministic, and unknown characters should be handled explicitly rather than silently discarded.

4. Data Preparation and Augmentation

A useful dataset contains images paired with accurate text transcriptions and, for detection, polygon or box annotations. The collection process should document the source, capture device, resolution, language variety, text type, and consent or licensing status. Personally identifiable information should be removed or protected when datasets contain real documents.

Preprocessing may include color conversion, resizing, normalization, and padding while preserving aspect ratio. Excessive resizing can erase tone marks, so minimum character height should be monitored. Augmentation should approximate plausible capture conditions rather than distort the writing beyond recognition.

- **Geometric:** small rotations, perspective transforms, and scale changes.
- **Photometric:** brightness, contrast, mild blur, and compression artifacts.
- **Occlusion:** limited shadows or partial obstruction, applied conservatively.
- **Background:** realistic texture variation that does not overwhelm the text.

Synthetic data can expand coverage of rare characters and fonts, but synthetic samples should not replace evaluation on genuine photographs. In particular, a model may learn rendering artifacts from a single font engine. A balanced dataset should include real examples across multiple sources and capture conditions.

5. Experimental Protocol

A credible OCR experiment should define its dataset split, preprocessing, vocabulary, model configuration, optimizer, training schedule, decoding method, and random seed. The test set should remain untouched during model selection. If several images originate from the same document or scene, they should be grouped into the same split to prevent near-duplicate leakage.

5.1 Detection metrics

Text detection can be evaluated using precision, recall, and an F-measure at a specified intersection-over-union threshold or using a benchmark's official polygon-matching protocol. The matching rule, ignored regions, and threshold must be stated because detection scores are not comparable when evaluation protocols differ.

5.2 Recognition metrics

Character Error Rate (CER) and Word Error Rate (WER) are useful complementary metrics. CER is computed from the minimum edit distance between predicted and reference character sequences, divided by the number of reference characters. WER applies the same principle to word tokens. The normalization policy—including whitespace, punctuation, and Unicode normalization—must be reported.

Metric	What it measures	Interpretation
Detection precision	Correctly detected text regions among predicted regions	High precision means fewer false text detections
Detection recall	Detected ground-truth regions among all ground-truth regions	High recall means fewer missed text regions
Character Error Rate	Character-level edit distance normalized by reference length	Lower is better; useful for diacritic-level errors
Word Error Rate	Word-level edit distance normalized by reference word count	Lower is better; reflects whole-word impact

6. Error Analysis

Aggregate scores can hide systematic failures. A Vietnamese OCR study should inspect substitutions between visually similar accented characters, missing tone marks, character merges, character splits, spacing errors, and punctuation mistakes. Errors should also be grouped by image condition, such as low resolution, motion blur, perspective distortion, and low contrast. This analysis helps determine whether the next improvement should target the detector, recognizer, preprocessing, or training data.

A confusion matrix over individual characters can reveal classes that are frequently confused, although sequence-level errors may involve insertions and deletions that a simple character confusion matrix does not capture. Representative failure cases should be documented with the original image crop, reference text, prediction, and a short explanation.

7. Reproducibility and Responsible Use

Reproducibility requires versioned annotations, a fixed vocabulary, documented preprocessing, recorded package versions, and saved model checkpoints. Training and evaluation scripts should be separated so that test data cannot accidentally influence training. A model card or concise technical note should state the intended use, known limitations, data sources, and evaluation conditions.

OCR outputs can contain sensitive personal information. Applications that process identity documents, invoices, or medical records should minimize data retention, restrict access, and avoid sending images to external services without an appropriate legal and privacy basis. Users should be able to correct recognized text, especially when OCR results feed into consequential decisions.

8. Limitations and Future Work

This paper provides a methodological framework and does not present a newly collected benchmark or measured model results. Its recommendations must be validated for the intended domain, such as street signs, printed forms, or receipts. Future work could compare a DB-based detector with alternative text detectors, evaluate CTC and attention-based recognizers under the same data split, and measure robustness to mobile-camera degradation.

9. Conclusion

Vietnamese OCR requires careful treatment of Unicode, diacritics, image quality, and evaluation design. A reproducible two-stage pipeline should make detection, recognition, data preparation, and error analysis independently testable. The most meaningful progress is likely to come not only from larger models, but also from better annotations, realistic data diversity, transparent metrics, and systematic analysis of character-level errors.

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Editorial note: This is an original methodological educational manuscript. It does not report newly conducted experiments or claim state-of-the-art performance. It has not undergone peer review.